



CHAPTER

Pretreatment and Filtration Skid

Disc filters, ultrafiltration unit and pretreatment piping

IN THIS CHAPTER

8.1	Overview and function	60
8.2	Key specifications	61
8.3	Operation	62
8.4	Maintenance	63
8.5	Troubleshooting	64

8.1 Overview and function

The pretreatment container conditions raw feed water before it reaches the SonixED™ stacks: disc filtration removes coarse suspended solids, and an ultrafiltration (UF) stage removes fine particulates and reduces fouling potential. Filtered water passes to the treatment container (chapter 3); rejected solids and backwash water are routed to the drain tank.



Figure 8.1. Suggested figure: photograph of the assembled pretreatment skid inside the container, with the disc filters, UF unit and main piping runs labeled.

8.2 Key specifications

- **Ultrafiltration:** Akualys/AKKIM APM 80 hollow-fiber UF module(s), 2" Victaulic connections. Two modules are confirmed ordered for HydroVolta (160 m² total membrane area). Membrane: modified PVDF hollow fiber, outside-to-inside flow, fiber OD/ID 1.4/0.8 mm, nominal MWCO ≤150,000 Da (≈30 nm pore size), 80 m² active area per module.
- **UF feedwater limits:** temperature ≤40 °C (typ. 25 °C); turbidity ≤250 NTU (typ. 50 NTU); TSS ≤80 mg/l; TOC ≤30 mg/l; pH 6–9; instantaneous free chlorine ≤150 mg/l.
- **UF operation:** dead-end/crossflow, filtrate flux 45–180 l/m²h at 25 °C, flow capacity 3.60–14.40 m³/h per module, feed inlet pressure 2–3 bar (instantaneous max. 5 bar), transmembrane pressure (TMP) 0.4–2 bar, filtrate SDI ≤2.5, filtrate turbidity ≤0.2 NTU.
- **Disc filtration:** AZUD Helix Automatic self-cleaning disc filters (FT-series, automatic backwash), 100-micron mesh discs; the documented BASF pilot P&ID additionally references disc-filter positions DF100/DF50/DF20 by micron rating — TODO confirm which AZUD Helix model number(s) map to the standard SalinBloc™ pretreatment train.
- Feed inlet: 2½" (M); filtered-water outlet: 2" (M); drain: 3" (M).
- Feed pump (pilot configuration): submersible, 4.5 m³/h at 6 bar.

TODO: confirm whether the AKKIM/AZUD models above are the standard SalinBloc™ bill-of-materials or pilot-specific; a finalized pretreatment BOM was not found. Several other filtration technologies (hydrocyclone separator, automatic screen filters, high-efficiency sand filters, microfilters) exist as vendor catalogs in HydroVolta's component library but are not confirmed as installed on any SonixED™ pretreatment train.

8.3 Operation

No dedicated pretreatment-specific operating procedure beyond the general startup/shutdown described in chapter 6 was found, but the UF module manufacturer documents the following normal operating cycle, which is the closest available reference for the standard SalinBloc™ pretreatment skid:

1. **Production:** a brief forward-flush step, then filtration (dead-end or crossflow) until the trigger for backwash is reached.
2. **Backwash:** drain, then air-scour (max. 20 Nm³/h at 600–800 mbar) to loosen the fouling layer, then backwash bottom-to-top and top-to-bottom (backwash pressure max. 2.5 bar) to flush it out.

TODO: confirm the actual backwash trigger used on SalinBloc™ (pressure differential vs. timer) and the cycle duration/frequency; the manufacturer spec above describes the module's capability, not HydroVolta's configured setpoint.

8.4 Maintenance

The UF manufacturer documents a Chemically Enhanced Backwash (CEB) / CIP cycle, triggered when TMP rises ≥ 0.9 bar above its initial value:

1. Run an extended backwash cycle (as in *Operation* above), then soak the module in cleaning chemical for part of the cycle.
2. CEB chemical concentrations: NaOCl 1000 mg/l, NaOH 500 mg/l, HCl 1000 mg/l, or citric acid 1–2% — select per the fouling type (organic/biological vs. scale) per chapter 12.
3. Total CIP duration ≈ 2 hours, at a cleaning flow of 1–2 m³/h per module.

For extended shutdowns (chapter 19), the manufacturer specifies a glycerol/sodium-metabisulfite preservation soak, and requires a minimum 4-hour flush with ≥ 14 m³ of feed water before restart. Before returning a preserved or newly installed module to service, disinfect with 100 ppm NaOCl circulated for 10 minutes, held for 1 hour, then drained until the filtrate runs chlorine-free. TODO: no document gives a routine cleaning-interval schedule (as opposed to the TMP-triggered CEB above) — see chapter 7 for the general log template.

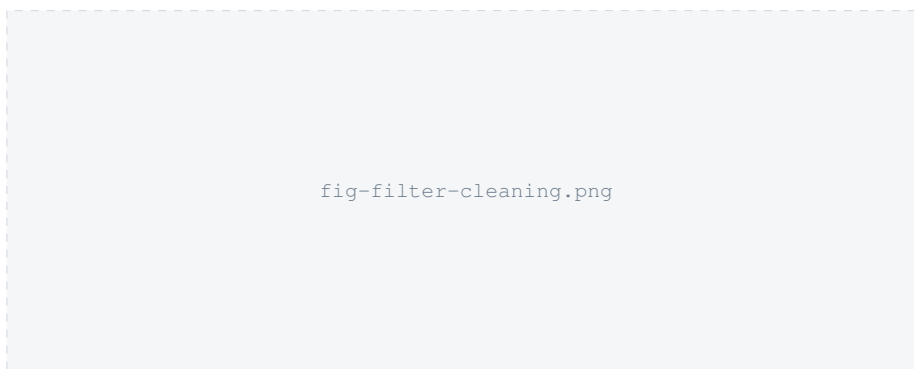


Figure 8.2. Suggested figure: step-by-step photos of the disc-filter backwash and UF CEB/CIP procedure above, once photographed on an actual unit.

8.5 Troubleshooting

- **Low filtered-water flow / high pressure drop:** check pump duty point against the 4.5 m³/h @ 6 bar rating and pipe sizing (60 mm feed line, 40 mm filtered-water line) before suspecting the filters themselves.
- **Rising transmembrane pressure (TMP):** if TMP has risen ≥ 0.9 bar above its initial value, run the CEB/CIP cycle above rather than waiting for a scheduled cleaning.
- **Poor downstream water quality:** verify feed water against the operating envelope in chapter 10 (turbidity, iron, SDI) and against the UF feedwater limits above — an out-of-envelope feed is a likely root cause on either stage.

TODO: no documented fault table (e.g. specific alarm codes) for the disc filters or UF unit was found beyond the operating-envelope figures above.